ROBERTO GUAGNELLI GARCÍA

Mechanical Engineering Student | Aeronautical Engineering Concentration

Mechanical design • Systems integration • Manufacturing & engineering analysis

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Graduation: July 2027 | Available: August 2027 | EU work authorization
Spanish citizen • English C2 • Certified SOLIDWORKS Professional (CSWP)

INDUSTRY EXPERIENCE

GE Aerospace — Querétaro, Mexico
Designer Intern, Product Definition | RISE — Aug 2026 – Present

- Designed **70+ support brackets** and completed **20+ harness-routing revisions** in Siemens NX for manufacturable installations, tool access and maintainability.
- Resolved clearance and minimum bend-radius issues; prepared CAD models and drawings using established datums and tolerance tables within the NX / Teamcenter workflow.
- Collaborate with vibration-analysis teams in **Mexico and the United States**, incorporating fatigue and vibration requirements into support-hardware and routing iterations.

Lean Summer Experience Intern | Integrated Performance & Operability — Jun – Jul 2026

- Developed planning tools, KPI dashboards and weekly reviews; meeting leaders recorded **approximately 20% shorter meetings**. Built Python automations for repetitive tasks.

AIRCRAFT DESIGN, INTEGRATION & LEADERSHIP

Borregos CEM AeroDesign | Tecnológico de Monterrey — Sep 2023 – Present
Engineering & Leadership Advisor — Sep 2026 – Present
Mentor the incoming Design Captain on design methods, technical reviews and cross-discipline integration.

Design Captain / Engineering Coordinator — Aug 2024 – Sep 2026

- Led **five designers** and coordinated interfaces across a **40+ member team**; modeled **300+ CAD parts per aircraft** in assemblies of 400+ components, from joints and landing gear to system installations.
- **Wingull:** Designed removable joints for a **4.2 m wing within a 1.2 m transport limit**; supported wing-load testing and integration of an aircraft that demonstrated payload-carrying flight.
- **Latías:** Built a **Python / XFLR5 longitudinal-dynamics model**; analyzed state-matrix eigenvalues and phugoid / short-period modes, then revised payload placement to recover modeled stability.
- Proposed and designed the Latías wing-load experiment to compare measured deflection with the structures team's prediction; coordinated aircraft integration through payload-carrying flight.

Manufacturing Coordinator | Hyperion — Sep 2023 – Aug 2024

- Coordinated **eight builders**; designed and built a spring-loaded aluminum nose gear and fitted the carbon-fiber fuselage skin. Team placed **2nd overall, SAE Aero Design México 2024**.

SELECTED TECHNICAL PROJECTS

- **Wing aerodynamics:** Developed steady / transient **ANSYS Fluent** simulations with SST k–ω, mesh refinement and XFLR5 comparisons to inform wing-incidence and tip-design decisions.
- **BIC extrusion cutter | Team project:** Contributed to a manufactured, instrumented prototype integrating a Geneva mechanism, gear train, feeder and cutter through a single motor drive.
- **Extruder failure investigation | Team project:** Combined hardness, metallography, SEM / EDS evidence and torsional calculations to assess a fractured H13 screw and competing failure hypotheses.

ADDITIONAL PROFESSIONAL EXPERIENCE

3D asset production & interactive development | Razzleberries and independent collaborations
Created **100+ Hello Kitty models and textures** to brand guidelines and **25+ Bloodborne models / variants**; developed creature behaviors and animation controllers for other Minecraft projects.

EDUCATION & TECHNICAL SKILLS

B.S. Mechanical Engineering | Tecnológico de Monterrey — Expected Jul 2027
Campus Estado de México | Aeronautical Engineering concentration | **91/100 average**

CAD & engineering: Siemens NX, Teamcenter, SOLIDWORKS (CSWP), engineering drawings, mechanical interfaces.
Analysis & programming: ANSYS Fluent, XFLR5, Python; CFD, flight dynamics, process automation.
Manufacturing: Composite fabrication, 3D printing, laser cutting, assembly and physical-test support.
Languages: Spanish (native), English (C2), German (A2).